

4(a)	transmissible ; A communicable I contagious pathogen ;	2
4(b)	bod abbreviations 1°, 2°, 3°, 4°, for the levels of protein structure any four from: <u>primary structure</u> (of each polypeptide) has, amino acid with sulfur in R-group / cysteine ; AW bod ref. to sequence of amino acids for primary structure <u>secondary structure</u> with, alpha helices / α-helices ; R beta-pleated sheets <u>tertiary structure</u>, qualified ; e.g. folded / coiled held in shape by R-group interactions / two named <i>idea of tertiary structure / active site / polypeptide chain A, has specific shape (complementary to substrate) ; A ref. to, binding site / specific catalytic site (in region of enzyme activity) instead of active site</i> <u>quaternary structure</u> has, two / 2, polypeptides / subunits ; further detail ; e.g. secondary structure with, areas of random coils quaternary structure disulfide bond / S–S (covalent bond), joins the two polypeptides (tertiary / quaternary) globular AVP ; e.g. zinc ion is, prosthetic group / cofactor A ref. to part of quaternary structure	4
4(c)	endocytosis ; bod phagocytosis	1

Question	Answer		Marks								
4(d)	<table><tr><th>description</th><th>types of immunity</th></tr><tr><td>Antibodies against tetanus toxin are passed from the mother to the fetus during pregnancy.</td><td>natural and passive ;</td></tr><tr><td>Inactivated tetanus toxin can be injected into a person.</td><td>artificial and active ;</td></tr><tr><td>Anti-tetanus antibodies are injected into a person who has been exposed to C. tetani.</td><td>artificial and passive ;</td></tr></table> A in either order for each example (i.e. 'passive and natural', 'active and artificial' and 'passive and artificial')	description	types of immunity	Antibodies against tetanus toxin are passed from the mother to the fetus during pregnancy.	natural and passive ;	Inactivated tetanus toxin can be injected into a person.	artificial and active ;	Anti-tetanus antibodies are injected into a person who has been exposed to C. tetani.	artificial and passive ;		3
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4(e)	F first <u>and</u> B second ; <i>max 1 mark if this mp not gained</i> C before A ; E before D ;		3								